

The background is a dark blue field filled with glowing elements. Numerous binary digits (0s and 1s) are scattered throughout, some appearing as vertical columns of code. Several keys are depicted in a glowing, cyan-blue color, some floating and others appearing to be part of the background pattern. The overall aesthetic is high-tech and digital.

KEY MANAGEMENT AND DISTRIBUTION

CHAPTER 14

Outlines

1. Symmetric-Key Distribution.
2. Kerberos.
3. Symmetric-Key Agreement.
4. Public-Key Distribution.
5. Public-Key Infrastructure.

Learning Objectives

In the end of this Chapter 14, students will be able:

1. To understand the importance and techniques of **symmetric-key distribution**.
2. To understand and describe briefly the **Kerberos** as an authentication server and key-distribution center.
3. To understand the technique of **symmetric-key agreement**.
4. To understand the importance and techniques of **public-key distribution**.
5. To understand the basic concept of **public-key infrastructure**.

SYMMETRIC-KEY DISTRIBUTION

Symmetric-Key Distribution – Overview

- Symmetric-key cryptography is generally **more efficient** than asymmetric-key cryptography for encrypting large messages.
- However, the challenging task in this symmetric-key cryptography is the **establishment and distribution of the shared secret key** between two communicating parties.

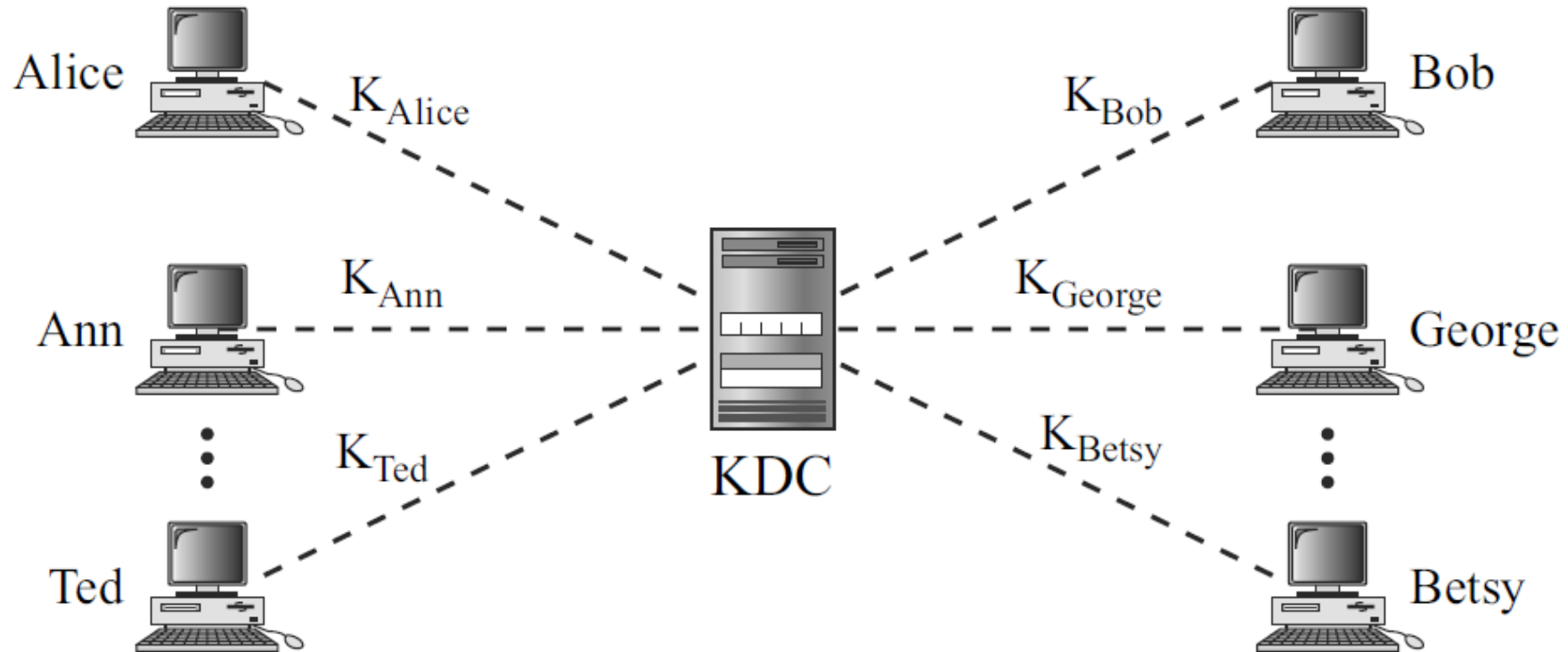
Symmetric-Key Distribution – Overview

- There are several ways that two parties Alice and Bob can achieve secret key distribution:
 1. Alice can select a key and physically deliver it to Bob.
 2. A third party can select a key and physically deliver it to Alice and Bob.
 3. If Alice and Bob have previously and recently used a secret key, either one can transmit to the other **a new secret key encrypted using the current key**.
 4. If Alice and Bob has an encrypted connection to a third party, then it can deliver a secret key on the encrypted links to both Alice and Bob.

Symmetric-Key Distribution – Key-Distribution Center (KDC)

- One practical solution is the use of Trusted Third Party (TTP), referred to as Key-Distribution Center (KDC).
- Each user establishes a shared secret key with the KDC to **reduce the number of keys needed**.
- For example, if Alice wants to send secret message to Bob, she will first send a request a session secret key from KDC for her and Bob. KDC relays such request to Bob and if Bob agrees, a **session key** is created for their secure communication. Such session key between Alice and Bob with KDC is used for authentication purpose too. (We will learn more about the **session key** in the later slides.)

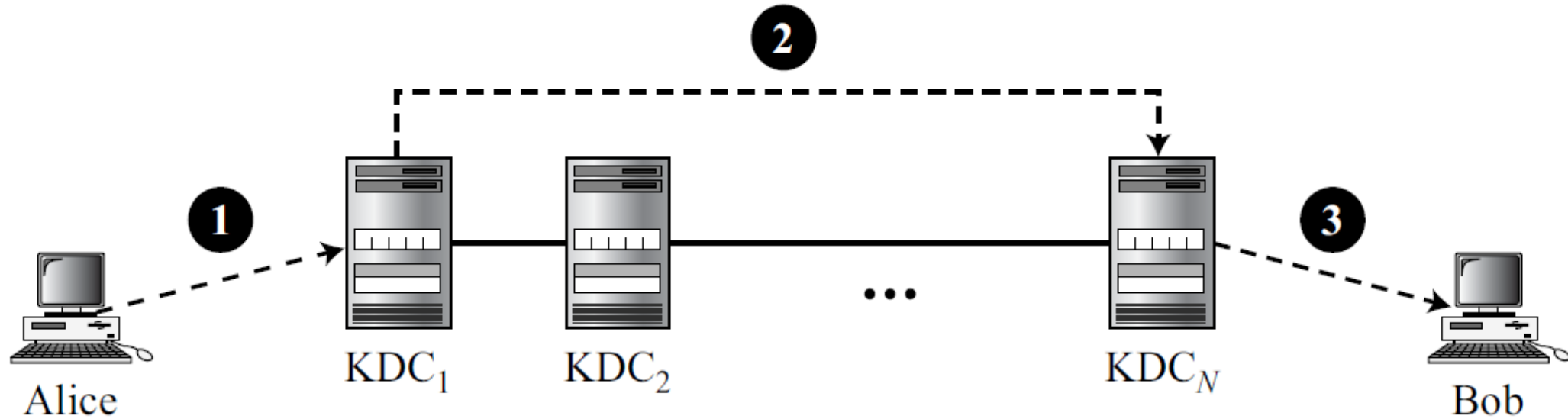
Symmetric-Key Distribution – Key-Distribution Center (KDC)



Symmetric-Key Distribution – Flat Multiple KDCs

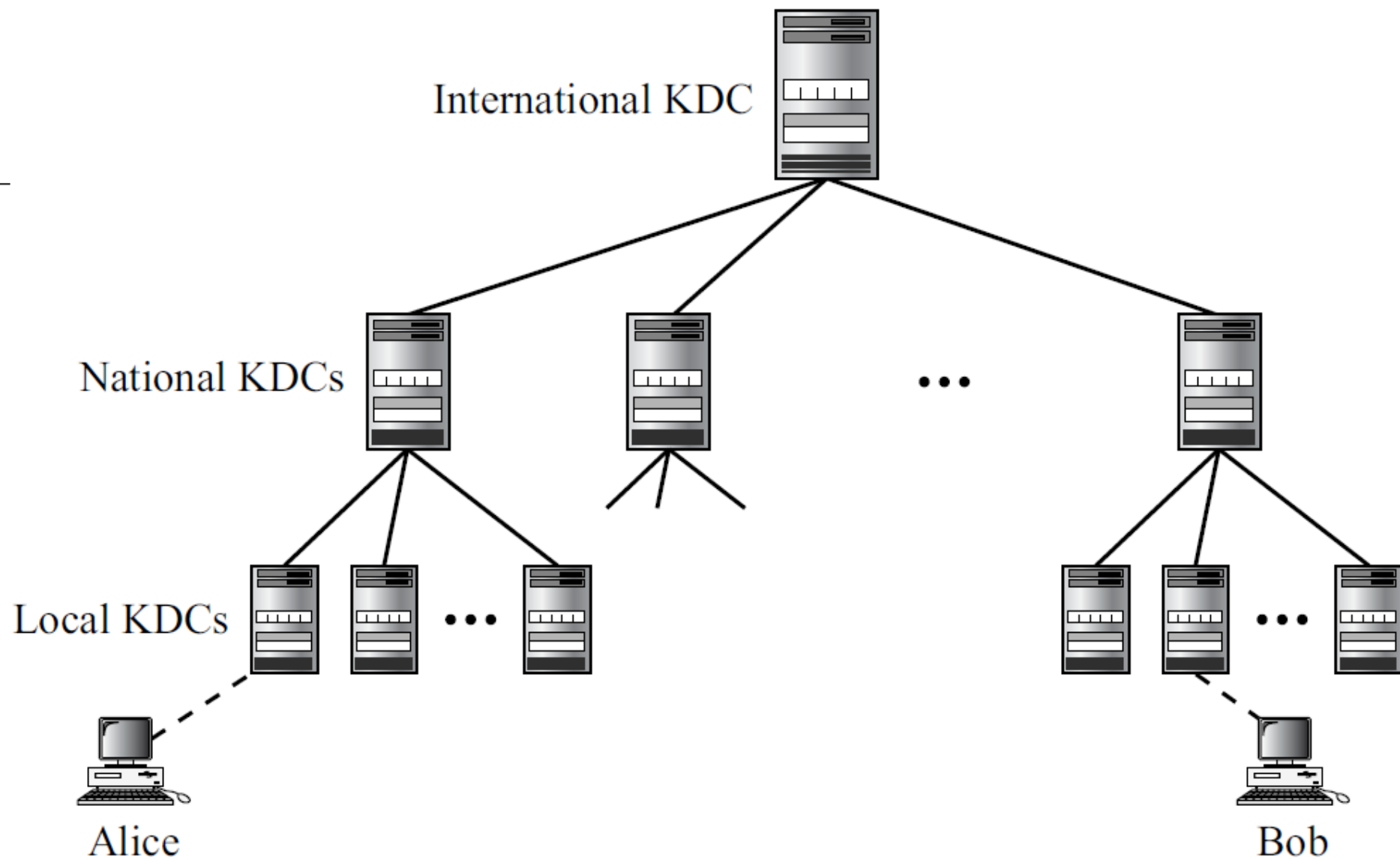
- When the number of users using a KDC increases, the system can become unmanageable and reach a bottleneck.
- To resolve this, a multiple KDCs can be established, and world can be divided into a few domains. Each domain can have one or more KDCs for redundancy in case of potential failure.
- One example of such multiple KDCs is the **Flat Multiple KDCs**, in which key-distribution is made possible for **users in different domains**.

Symmetric-Key Distribution – Flat Multiple KDCs



Symmetric-Key Distribution – Hierarchical Multiple KDCs

- The concept of Flat Multiple KDCs can be extended to a hierarchical system of KDCs, with one or more KDCs on top of the hierarchy. For instance, local KDCs, national KDCs, and international KDCs.
- The request is made in such a way that it relays level by level until it reaches the desired destination.



Symmetric-Key Distribution – Session Keys

- When KDC creates a secret key for each user, the **key is used between KDC and users, NOT between users themselves.**
- For users to establish their shared secret key, KDC creates a **session key** for such a purpose upon request. This **session can be used only once**, and **it becomes useless when the communication is terminated.**
- There are several approaches in which session keys can be established.
 1. **Simple (Basic) Protocol using a KDC.** (We only focus on this in this course)
 2. Needham-Schroeder Protocol.
 3. Otway-Rees Protocol.

Session Keys via Simple Protocol

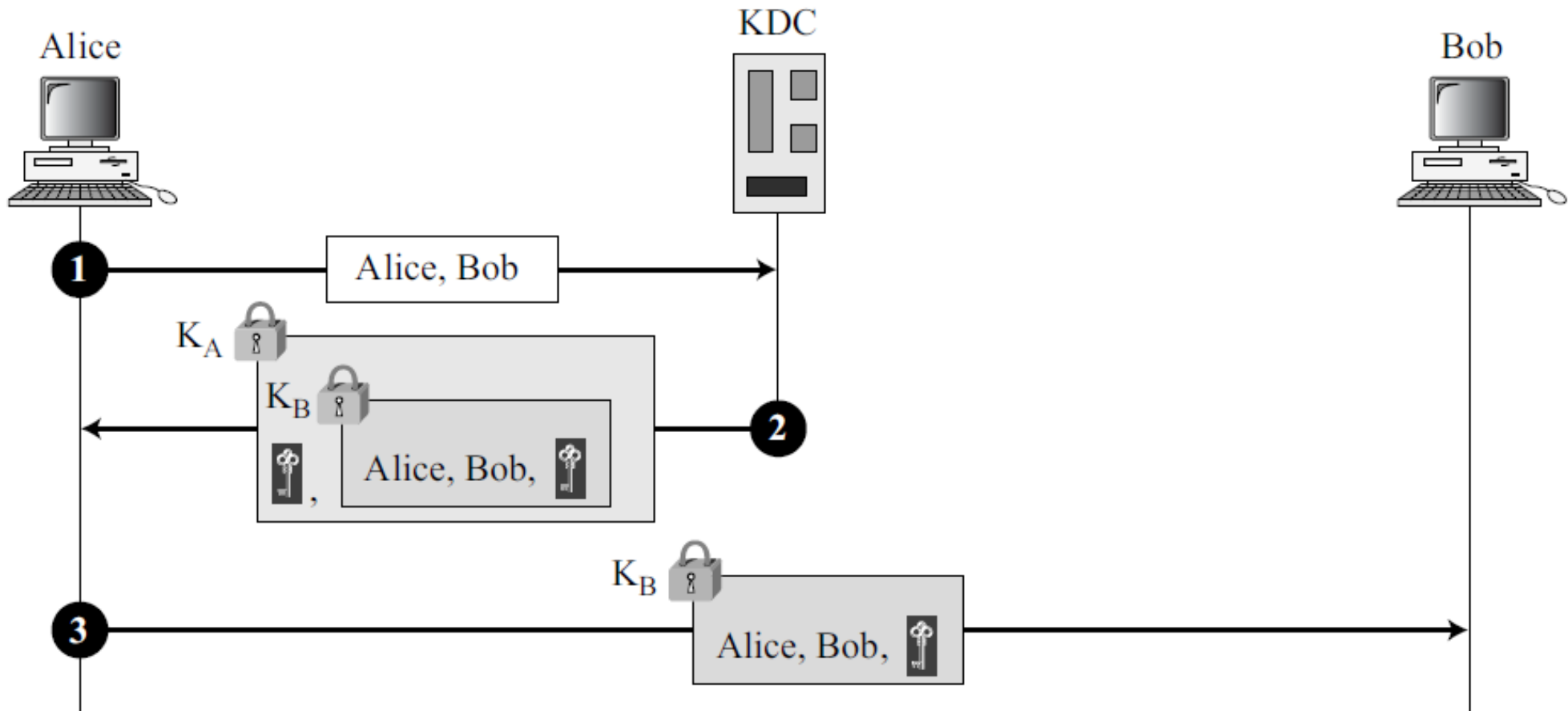
1. Alice sends a plaintext message to the KDC to obtain a symmetric session key between Bob and herself. The message contains her registered ID_A and Bob's ID_B . This message is not encrypted, and it is public.
2. The KDC receives the message and creates **a ticket** encrypted using Bob's key K_B . The ticket contains (ID_A, ID_B) and the session key K_{AB} . The ticket with a copy of the session key is sent to Alice. Alice receives the message, decrypts it, and extracts the session key K_{AB} . She cannot decrypt Bob's ticket.
3. Alice sends the ticket to Bob. Bob decrypts the ticket and knows that Alice needs to send messages to him using K_{AB} as the session key.

K_A  Encrypted with Alice-KDC secret key

 Session key between Alice and Bob

K_B  Encrypted with Bob-KDC secret key

KDC: Key-distribution center



Session Keys via Simple Protocol

- **Notes:**

- In Step 2, the message contains **a double encryption**; the ticket is encrypted, and the entire message is also encrypted. In the second message, **Alice is authenticated to the KDC**, because only Alice can open the whole message using her secret key with KDC.
- In Step 3, **Bob is authenticated to the KDC** as only Bob can open the ticket. Since Bob is authenticated to the KDC, **he is also authenticated to Alice**, who trusts the KDC. In the same way, **Alice is also authenticated to Bob**, because Bob trusts the KDC and the KDC has sent Bob the ticket that includes the identity of Alice.

KERBEROS

Kerberos

- Kerberos is an **authentication protocol** at the same time **a KDC**. It is popularly used in the Windows 2000.
- It is originally designed at MIT and has gone through several versions.
- In this course, we consider only the Version 4.

Kerberos (Overview)

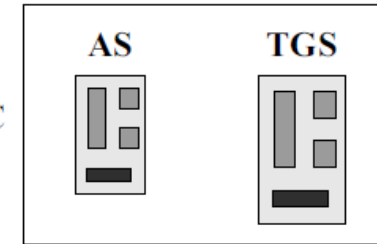
AS: Authentication server

TGS: Ticket-granting server

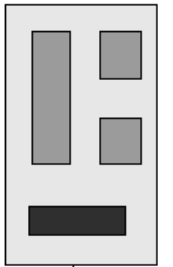
User (Alice)



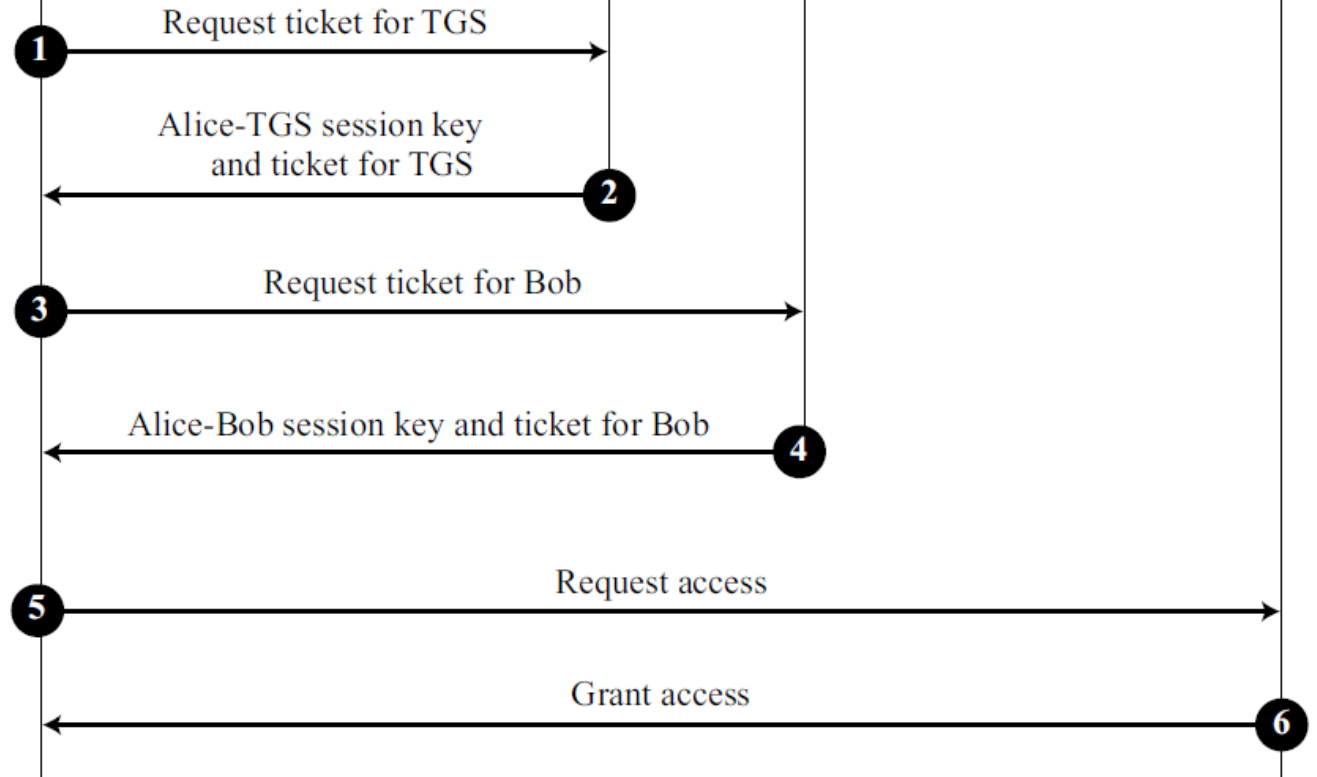
KDC



Server (Bob)

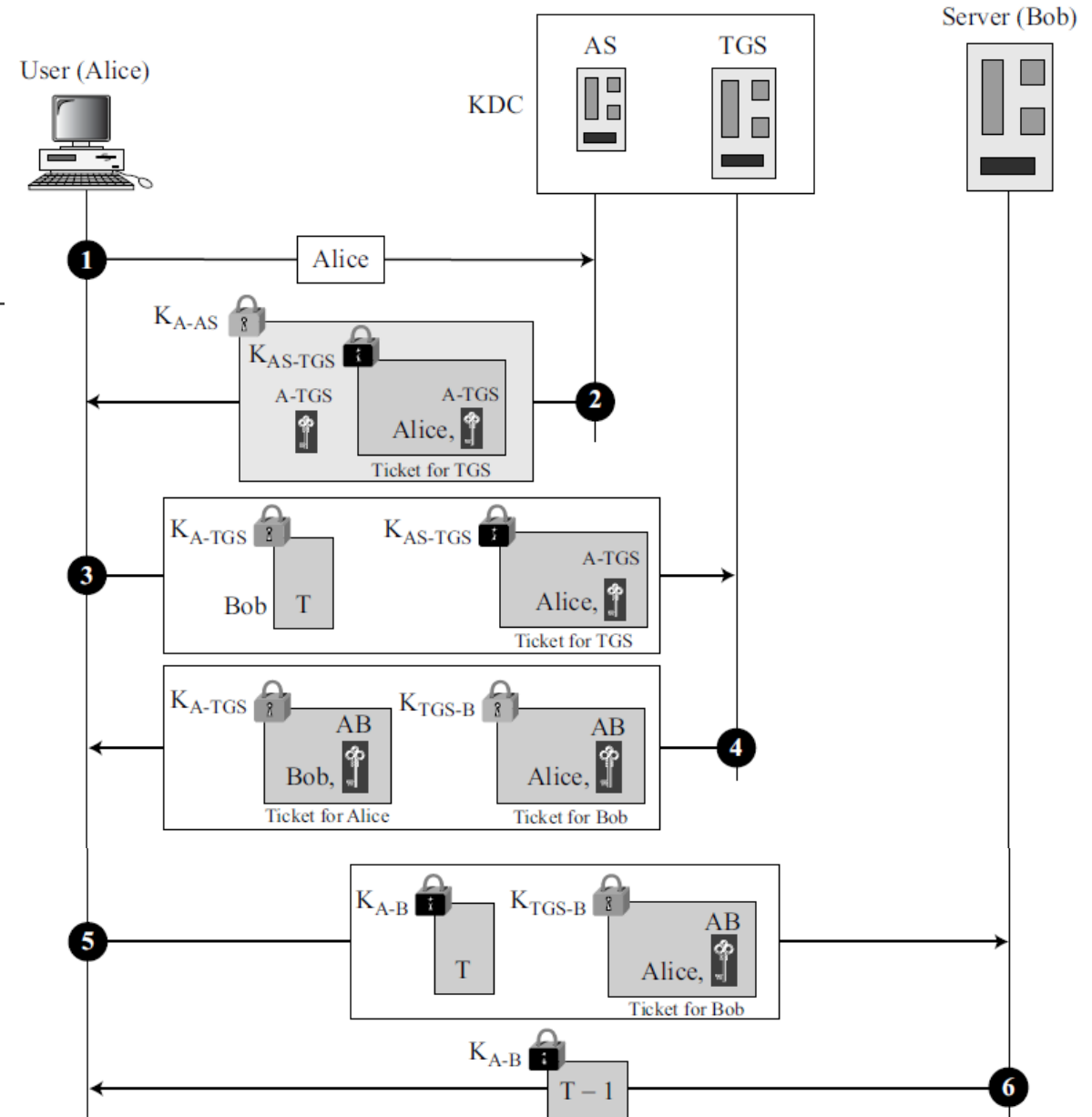
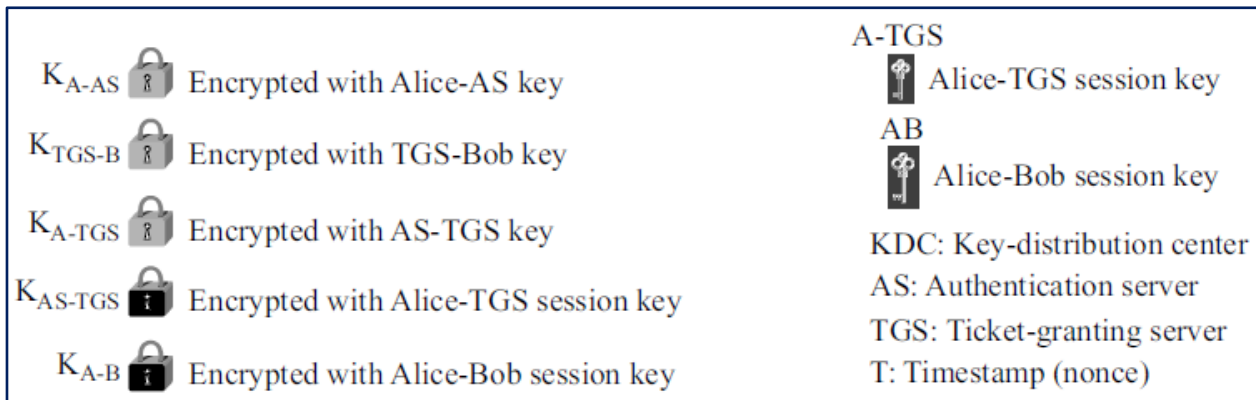


- The Kerberos servers.
- **AS:** This is the KDC in Kerberos protocol.
- **TGS:** It issues a ticket for the real server (service provider).
- **Real Server:** The service provider for requested user.



Kerberos (Example)

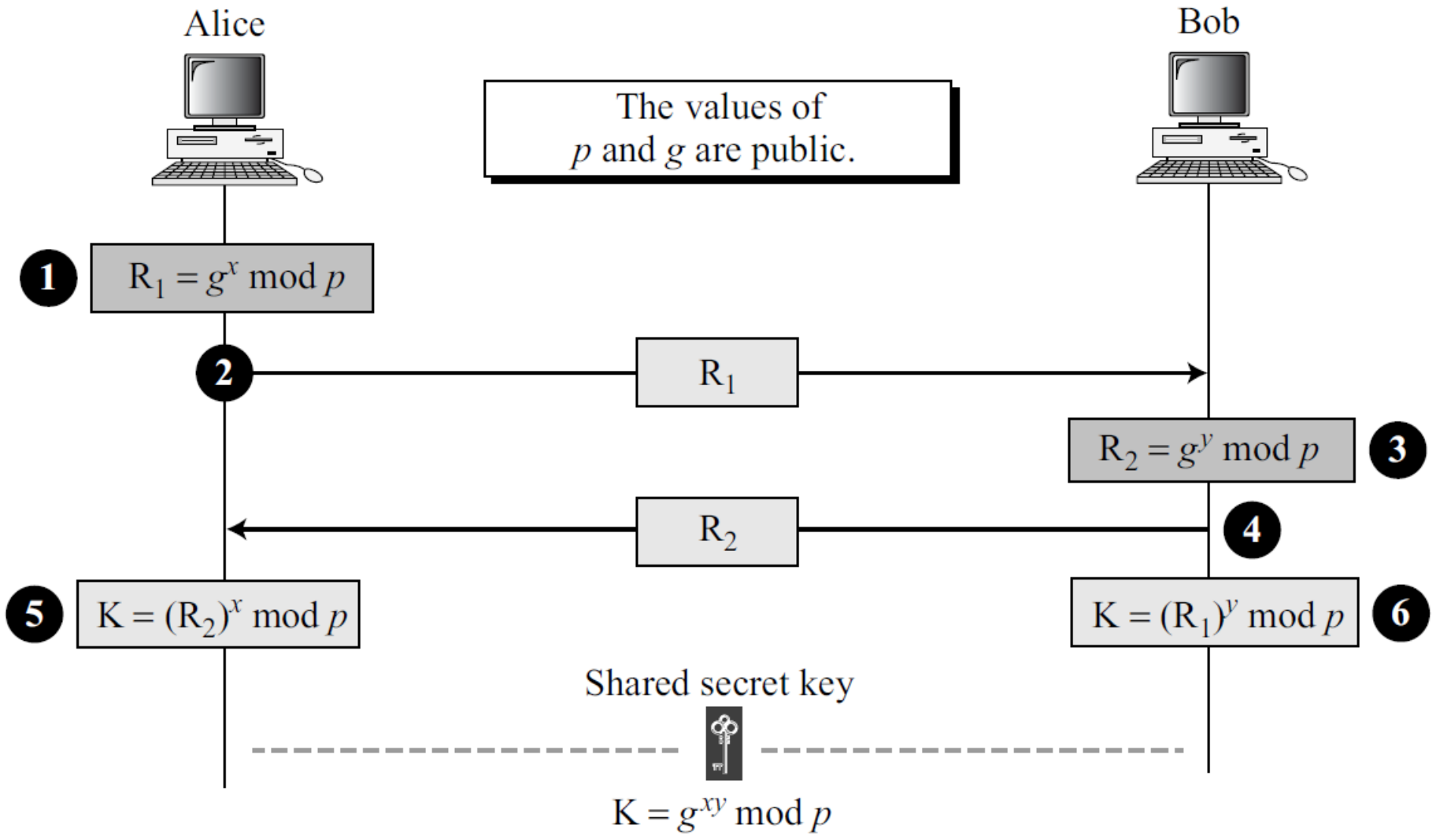
- Can you explain verbally (or write in the algorithmic flow) on how the Kerberos works as authenticating server and KDC?



SYMMETRIC-KEY AGREEMENT

Symmetric-Key Agreement

- It is possible for Alice and Bob to establish a session key **without needing the help of a KDC**. Such an establishment is referred to as **symmetric-key agreement**.
- The most common method for such an objective is the **Diffie-Hellman Key Exchange**.
- Since we have covered the DHKE completely in **Chapter 9**, you are advised to refer to the notes for another revision.
- In the next slide, we outline the general framework once again.



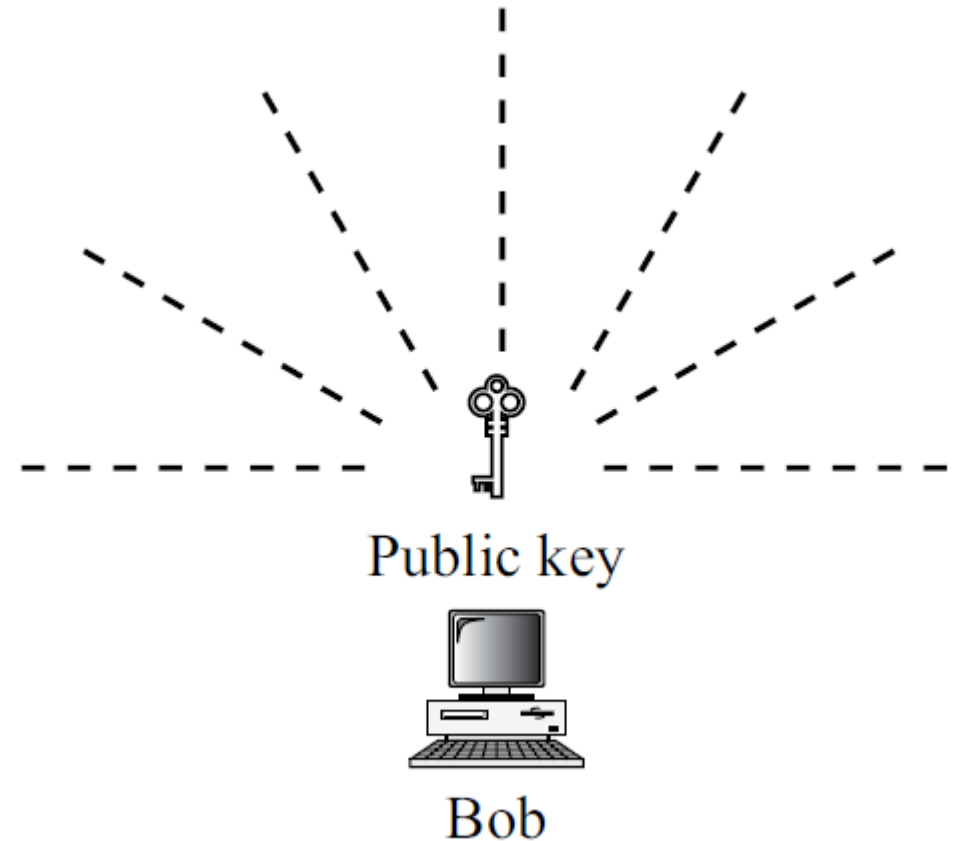
PUBLIC-KEY DISTRIBUTION

Public-Key Distribution

- In public-key cryptography, users do not have to share a symmetric key.
- Since every user has his/her own pair of **public-private keys**, message transmission can be executed immediately using the **public keys**.
- As **public keys** are available to any users, **distributing the keys** must be done accordingly for them to be useful. There are several ways that this can be done:
 1. Public Announcement.
 2. Trusted Center.
 3. Controlled Trusted Center.
 4. Certification Authority (CA).

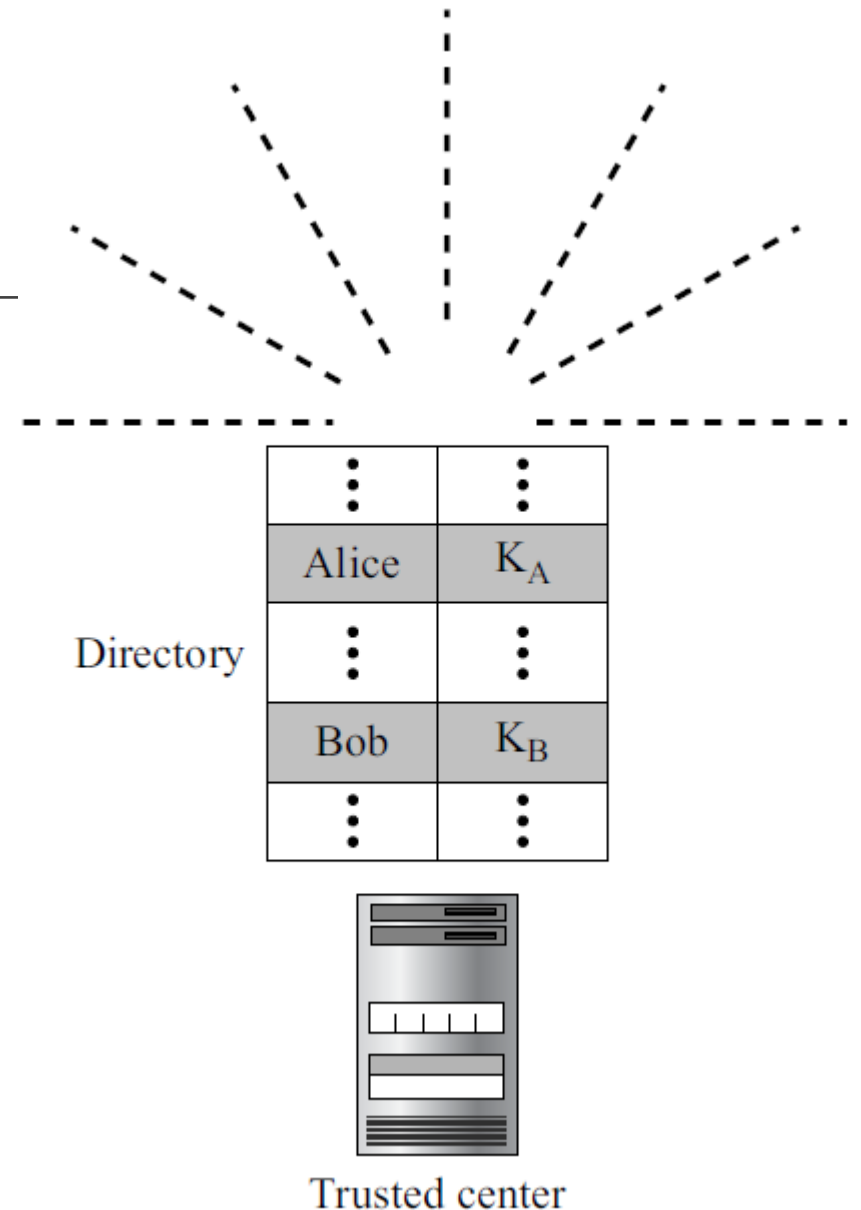
Public-Key Distribution – Public Announcement

- The naïve approach in which users publicly announcement their public keys on the website, or local newspapers (or any media that public can access easily).
- The approach is **insecure** since **forgery** is possible by an adversary.



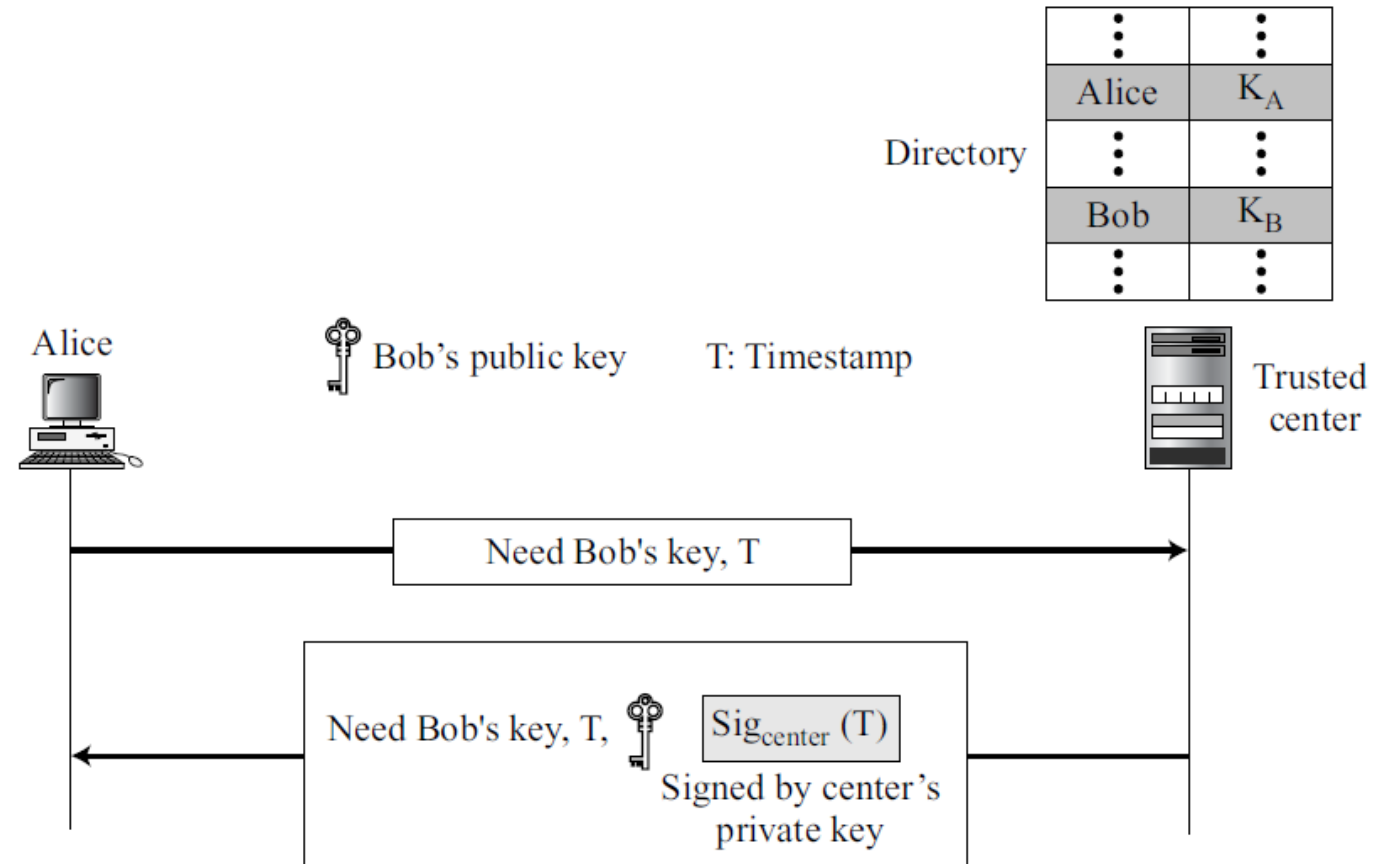
Public-Key Distribution – Trusted Center

- A more secure approach where a trusted center retains a public directory of all the public keys and dynamically updates it regularly.
- Users can submit their public key to the trusted center and keep their private securely with them.
- Such approach will require the users to register themselves for identification purpose in the future.



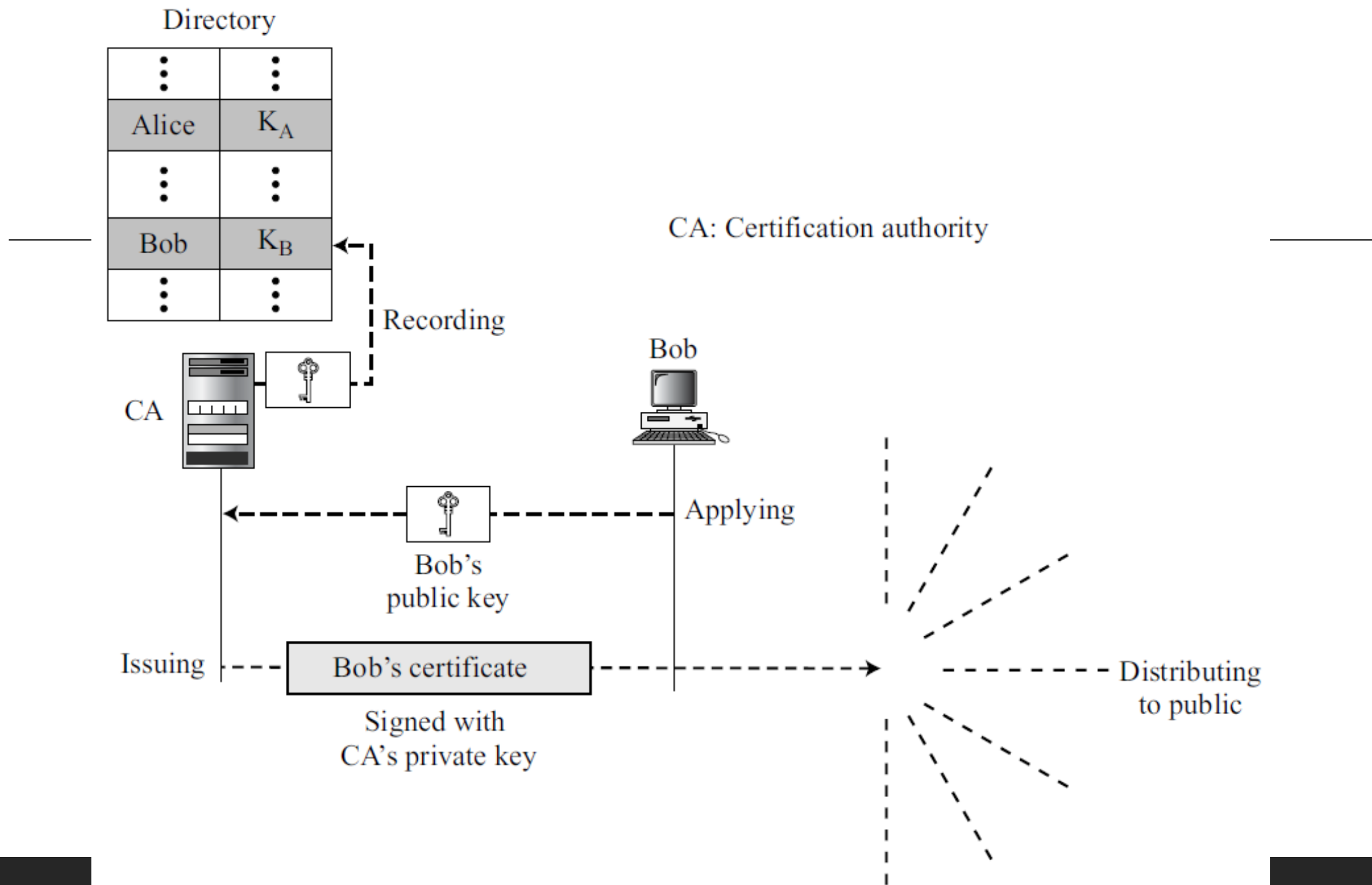
Public-Key Distribution – Controlled Trusted Center

- A higher level of security compared to trusted center.
- The public key announcements can include timestamp and signed by the authority to prevent interception and modification of the response.



Public-Key Distribution – Certification Authority

- All previous approaches can result heavy load on the center if the number of public key requests is large.
- An alternative solution is to create **public-key certificates** which involves a **certificate authority (CA)** that **binds a public key to the user and issues a certificate**.
- This approach can prevent users' public keys from being forged since **any submitted public keys for certification will be signed by CA using CA's private key**.



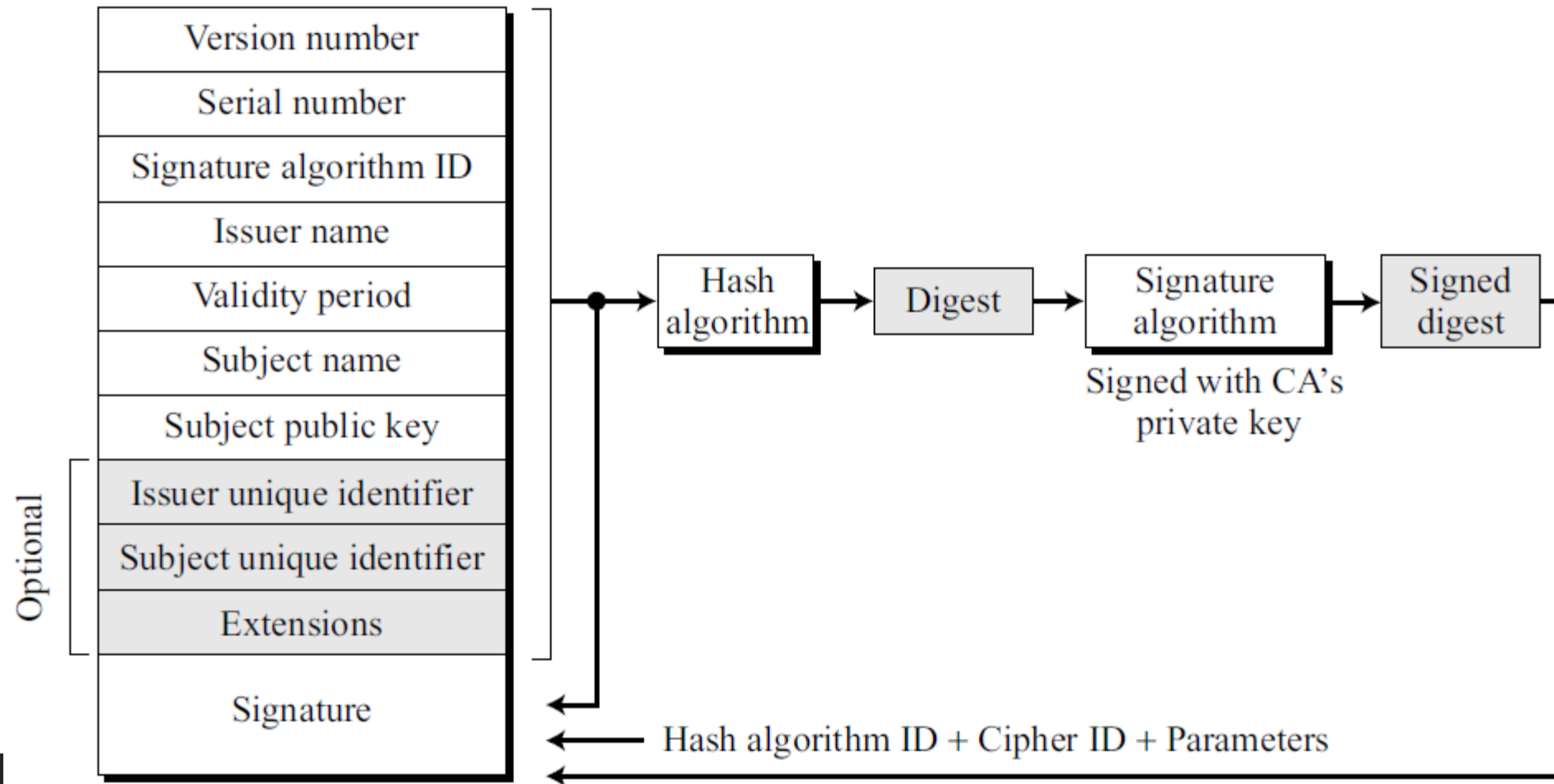
PUBLIC-KEY DISTRIBUTION

X.509 CERTIFICATION

Public-Key Distribution – X.509 Certification

- Although the presence of CA resolves the public-key distribution issue, however it contains some unavoidable side effect, since different users may have different format in their certificate.
- A standardization needs to be done to make the certificate universal. Hence, the **X.509 is designed** by ITU to perform such task.

Public-Key Distribution – X.509 Certificate Format



Public-Key Distribution – X.509 Certification

- **Certificate Renewal**

- Each certificate has a period of validity. If there is no problem with the certificate, the CA issues a new certificate before the old one expires.

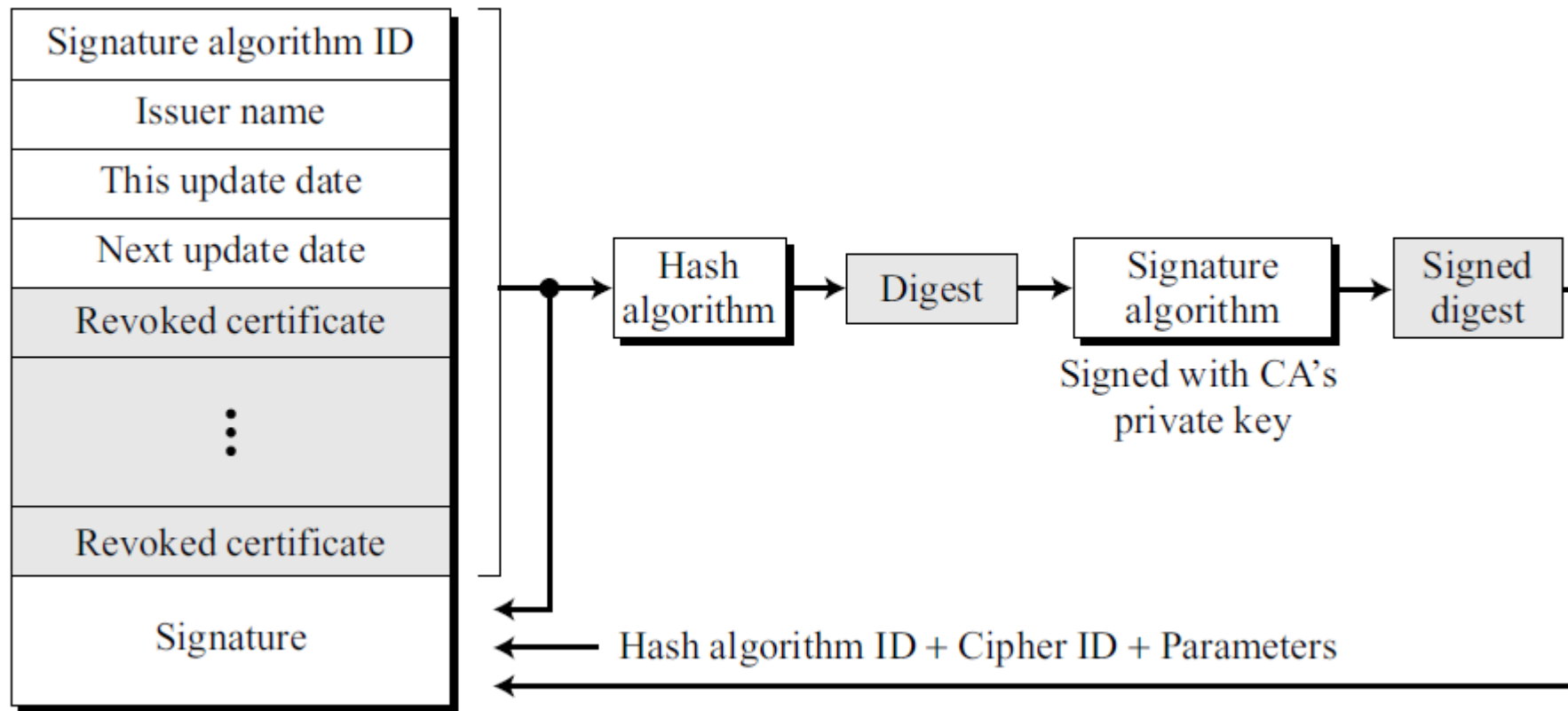
- **Certificate Revocation**

- In some cases, a certificate must be revoked before its expiration.

- **Delta Revocation**

- To make revocation more efficient, the delta certificate revocation list (delta CRL) has been introduced.

Public-Key Distribution – X.509 Certificate Revocation Format

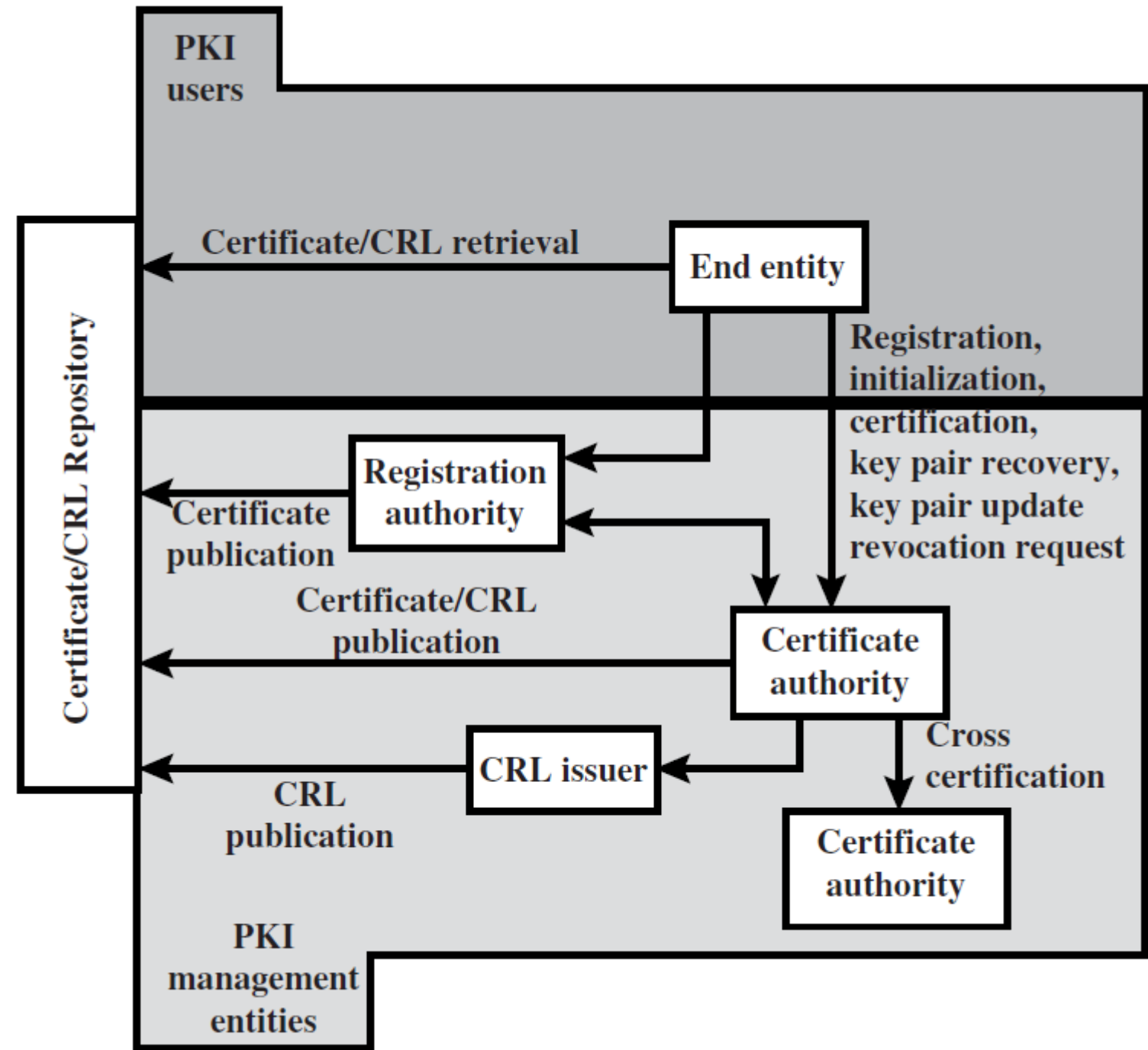


PUBLIC-KEY INFRASTRUCTURE (PKI)

Public-Key Infrastructure (PKI)

- Public-Key Infrastructure (PKI) is a model for creating, distributing, and revoking certificates based on the X.509.
- The Internet Engineering Task Force (IETF) has created the **PKI X.509 (PKIX)**.
- There are **several duties** have been defined for a PKI:
 1. Certificates' issuing, renewal, and revocation.
 2. Keys' storage and update.
 3. Providing services to other protocols.
 4. Providing access control.

Public-Key Infrastructure X.509 (PKIX) Architectural Model



Main References for Chapter 14

1. Forouzan, B.A. “*Introduction to Cryptography and Network Security*”, Third Edition, McGraw-Hill, 2008.
2. Stallings, W. “*Cryptography and Network Security: Principles and Practices*”, Fifth Edition, Pearson Prentice Hall, 2011.

Way Forward...

Types Public-Key Cryptography

1. Encryption scheme
2. Key exchange mechanism
3. Key encapsulation mechanism
4. Digital signature scheme
5. Identification scheme
6. Identity-Based (ID) cryptography
7. Certificateless cryptography
8. Elliptic curve cryptography
9. Signcryption scheme
10. Hybrid cryptography
11. Pairing-Based cryptography
12. Quantum cryptography
13. Post-Quantum cryptography

Quantum Computing Era

Quantum Cryptanalysis

Shor's Algorithm

Polynomial-Time Algorithms for Prime Factorization and Discrete Logarithms on a Quantum Computer*

Peter W. Shor[†]

Abstract

A digital computer is generally believed to be an efficient universal computing device; that is, it is believed able to simulate any physical computing device with an increase in computation time by at most a polynomial factor. This may not be true when quantum mechanics is taken into consideration. This paper considers factoring integers and finding discrete logarithms, two problems which are generally thought to be hard on a classical computer and which have been used as the basis of several proposed cryptosystems. Efficient randomized algorithms are given for these two problems on a hypothetical quantum computer. These algorithms take a number of steps polynomial in the input size, e.g., the number of digits of the integer to be factored.

Keywords: algorithmic number theory, prime factorization, discrete logarithms, Church's thesis, quantum computers, foundations of quantum mechanics, spin systems, Fourier transforms

*A preliminary version of this paper appeared in the Proceedings of the 35th Annual Symposium on Foundations of Computer Science, Santa Fe, NM, Nov. 20–22, 1994, IEEE Computer Society Press, pp. 124–134.

[†]AT&T Research, Room 2D-149, 600 Mountain Ave., Murray Hill, NJ 07974.

Quantum Cryptanalysis

Grover's Algorithm

A fast quantum mechanical algorithm for database search

Lov K. Grover
3C-404A, AT&T Bell Labs
600 Mountain Avenue
Murray Hill NJ 07974
lkg@mhcnet.att.com

Summary

An unsorted database contains N records, of which just one satisfies a particular property. The problem is to identify that one record. Any classical algorithm, deterministic or probabilistic, will clearly take $O(N)$ steps since on the average it will have to examine a large fraction of the N records. Quantum mechanical systems can do several operations simultaneously due to their wave like properties. This paper gives an $O(\sqrt{N})$ step quantum mechanical algorithm for identifying that record. It is within a constant factor of the fastest possible quantum mechanical algorithm.

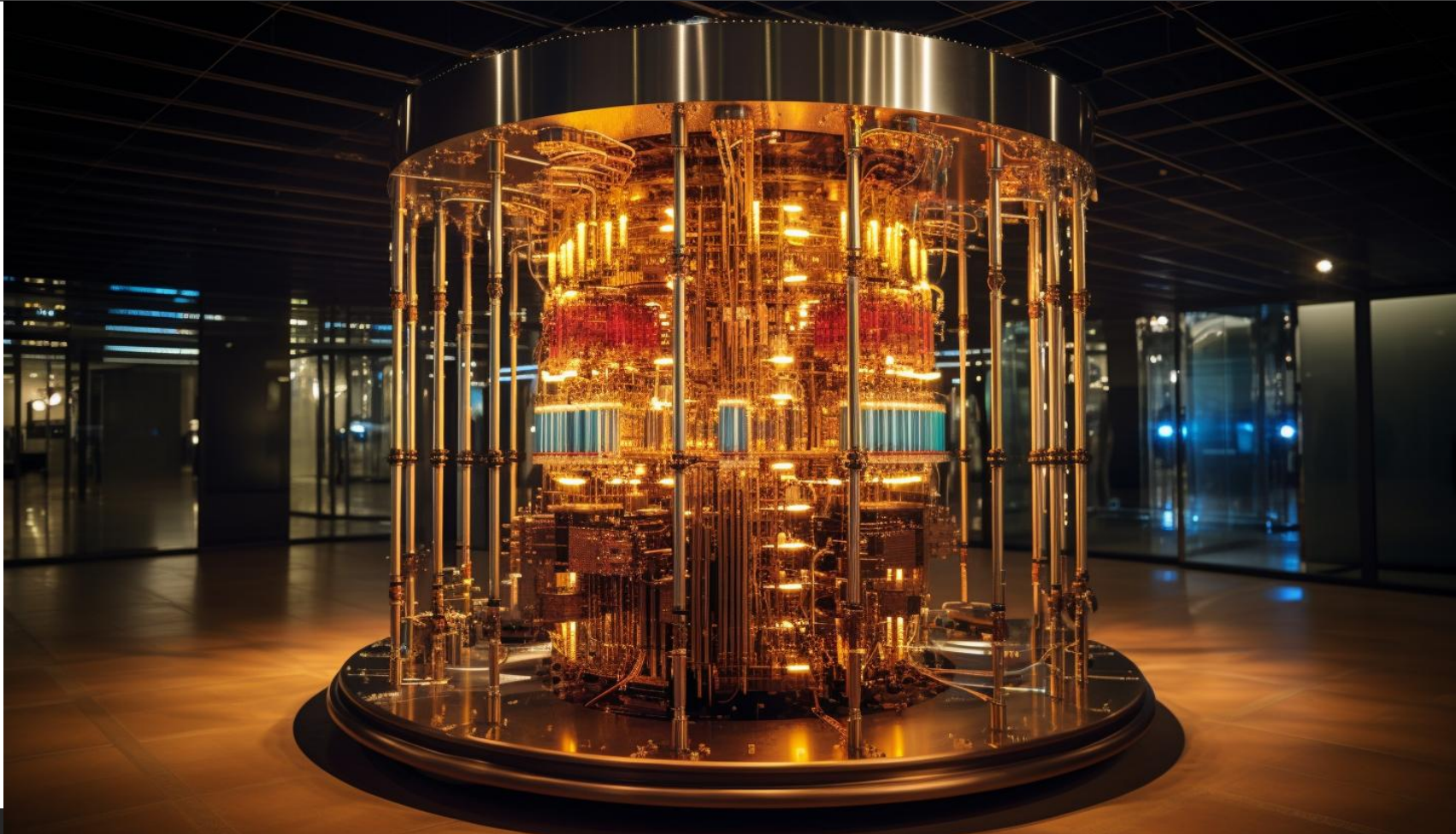
This paper applies quantum computing to a mundane problem in information processing and presents an algorithm that is significantly faster than any classical algorithm can be. The problem is this: there is an unsorted database containing N items out of which just one item satisfies a given condition - that one item has to be retrieved. Once an item is examined, it is possible to tell whether or not it satisfies the condition in one step. However, there does not exist any sorting on the database that would aid its selection. The most efficient classical algorithm for this is to examine the items in the database one by one. If an item satisfies the required condition stop; if it does not, keep track of this item so that it is not examined again. It is easily seen

Permission to make digital/hard copies of all or part of this material for personal or classroom use is granted without fee provided that the copies are not made or distributed for profit or commercial advantage, the copyright notice, the title of the publication and its date appear, and notice is given that copyright is by permission of the ACM, Inc. To copy otherwise, to republish, to post on servers or to redistribute to lists, requires specific permission and/or fee.

STOC'96, Philadelphia PA, USA

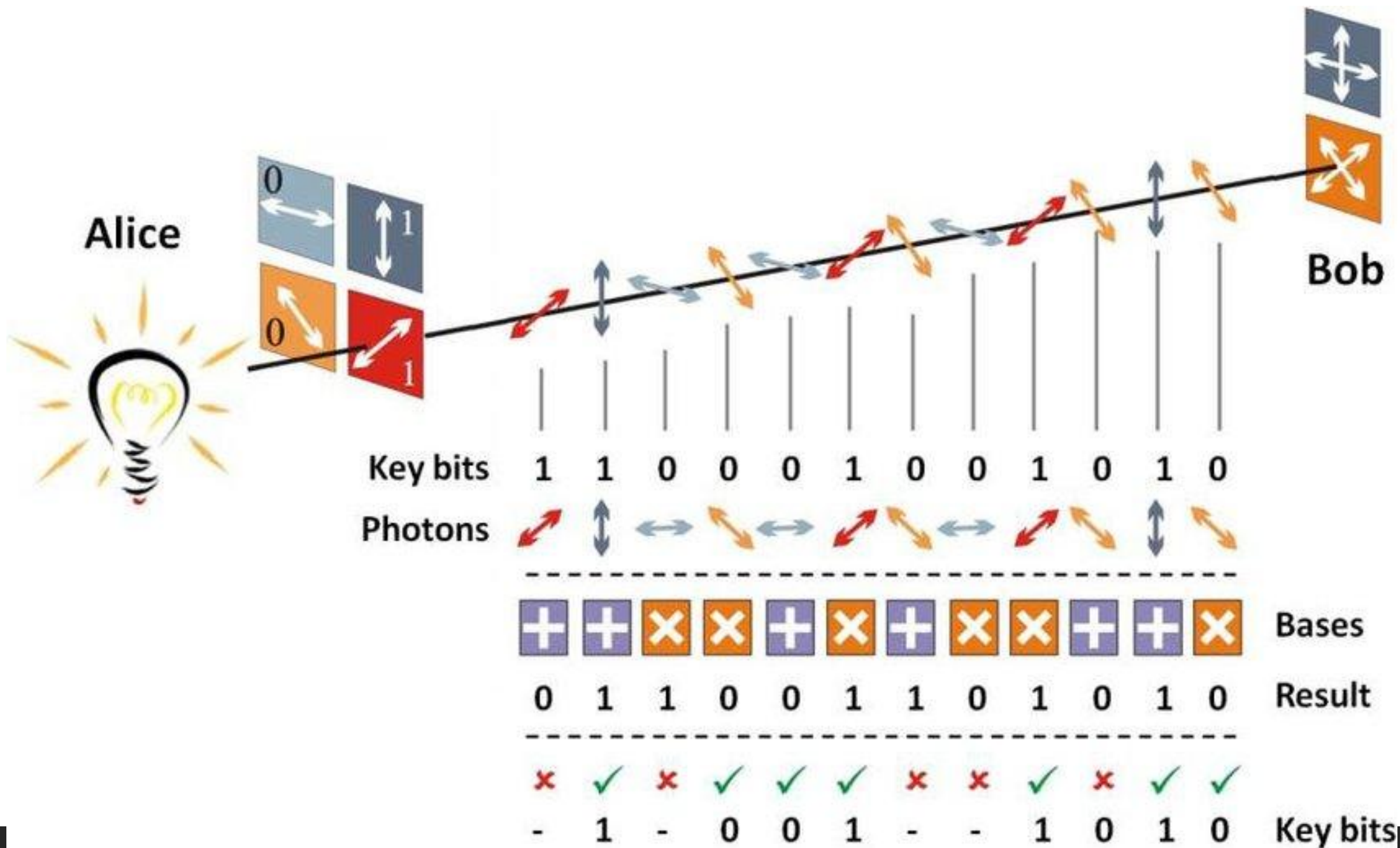
© 1996 ACM 0-89791-785-5/96/05..\$3.50

Quantum Cryptography



Quantum Cryptography

- **Quantum cryptography** (also known as **quantum encryption**) refers to various cybersecurity methods for encrypting and transmitting secure data based on the naturally occurring and immutable laws of quantum mechanics (physics).
- One of the famous quantum cryptography is the **Quantum Key Distribution (QKD) – Bennett and Brassard scheme (BB84)**.



Post-Quantum Cryptography

Post-Quantum Cryptography

- Cryptographic systems that are **secure against both quantum and classical computers and** can interoperate with existing communications protocols and networks.

Post-Quantum Cryptography Candidates (Primitives)

1. Hash-based cryptography
2. Code-based
3. Lattice-based
4. Multivariate Quadratic Polynomial
5. Elliptic-curve Isogeny-based

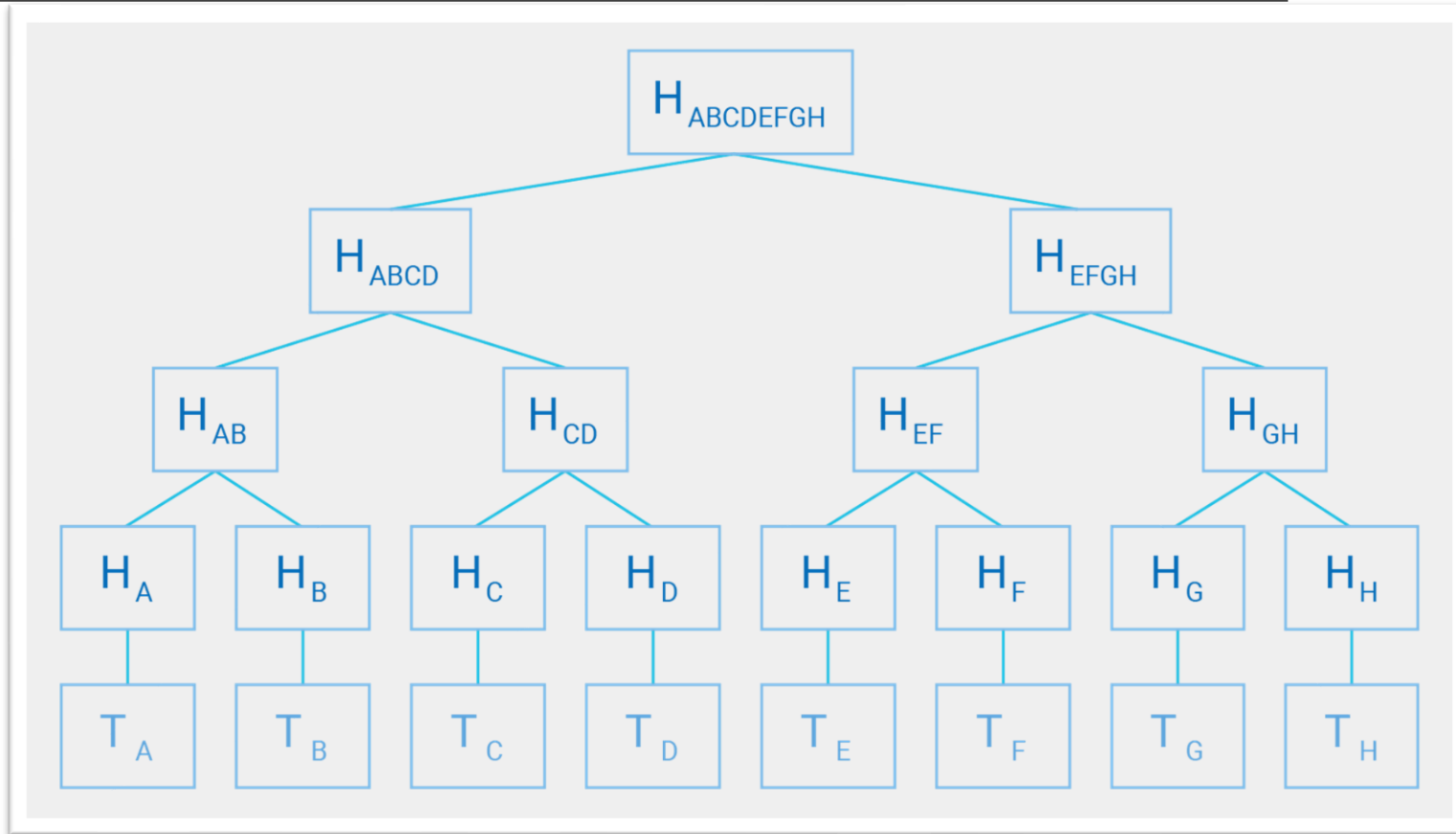
Hash-based Cryptography

- The design of cryptography based on **hash function**.
- The underlying security lies on the difficulty of finding:

1. Collision

2. Pre-Image

3. Second Pre-Image



Code-based Cryptography

- The design of cryptography based on the coding theory – **the error-correcting codes**.
- The Code-based Cryptography is closely related to **lattices** too.
- The famous code-based cryptographic scheme – The McEliece cryptosystem.

Lattice-based Cryptography

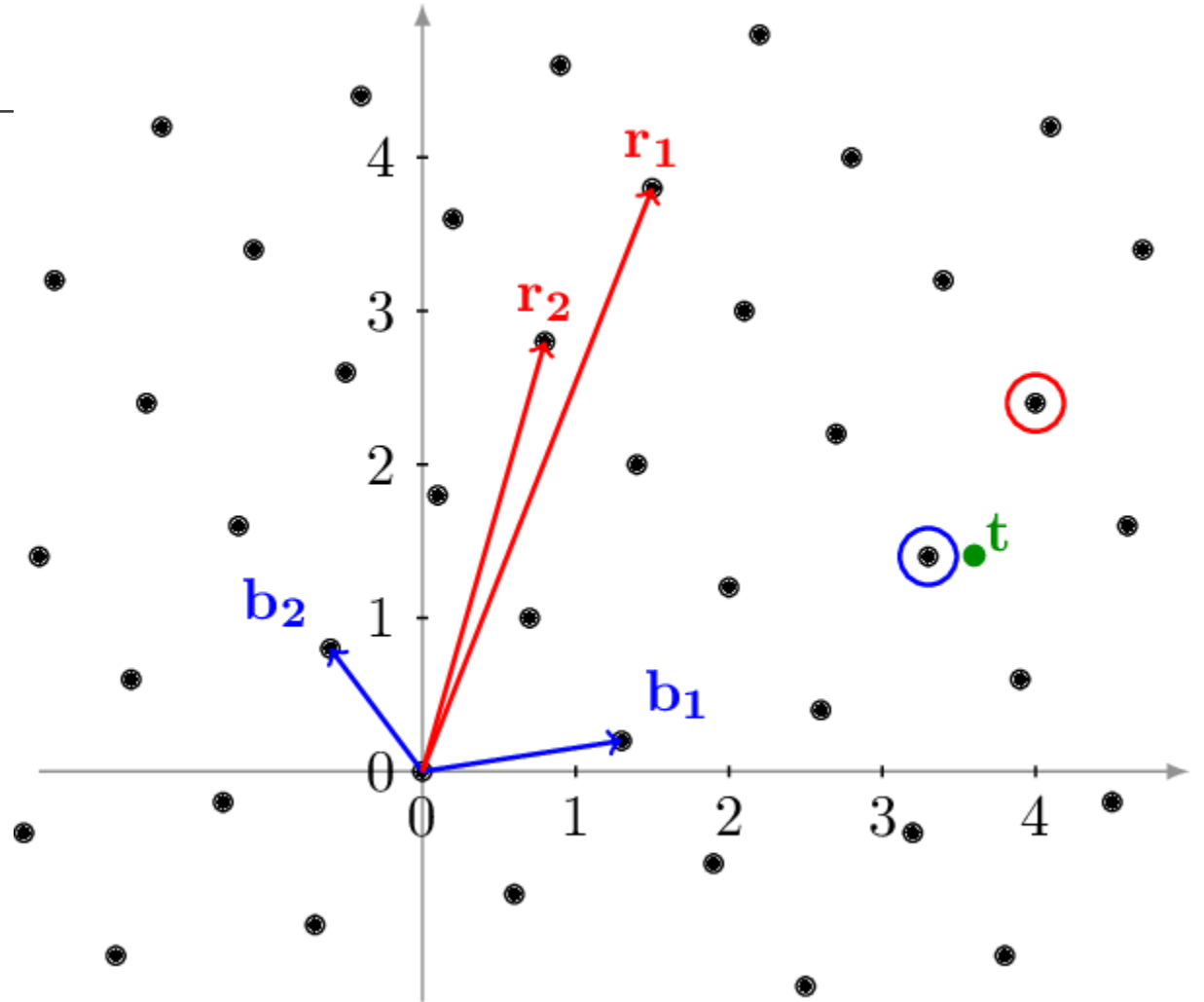
➤ The design of cryptography based on **lattice**.

➤ The underlying security lies on the difficulty of solving:

1. **Shortest Vector Problem (SVP)**

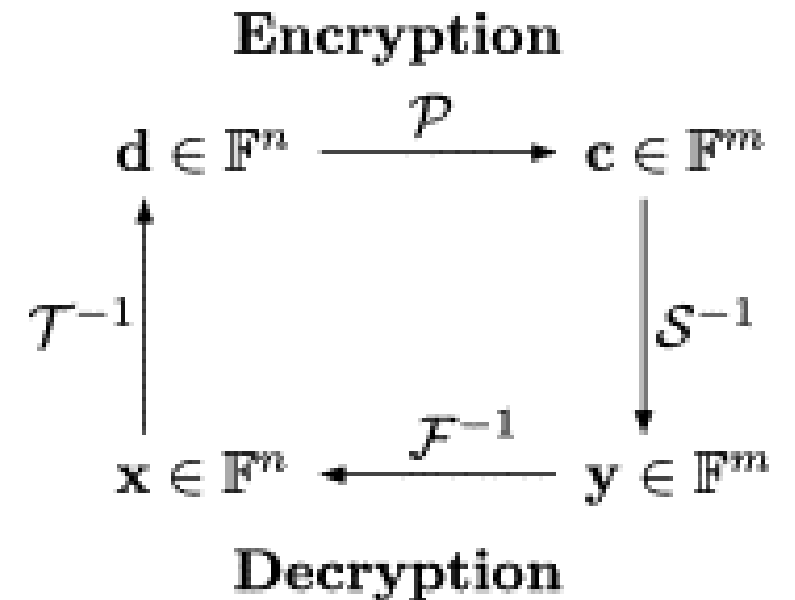
2. **Closest Vector Problem (CVP)**

3. **Learning with Errors (LWE)**



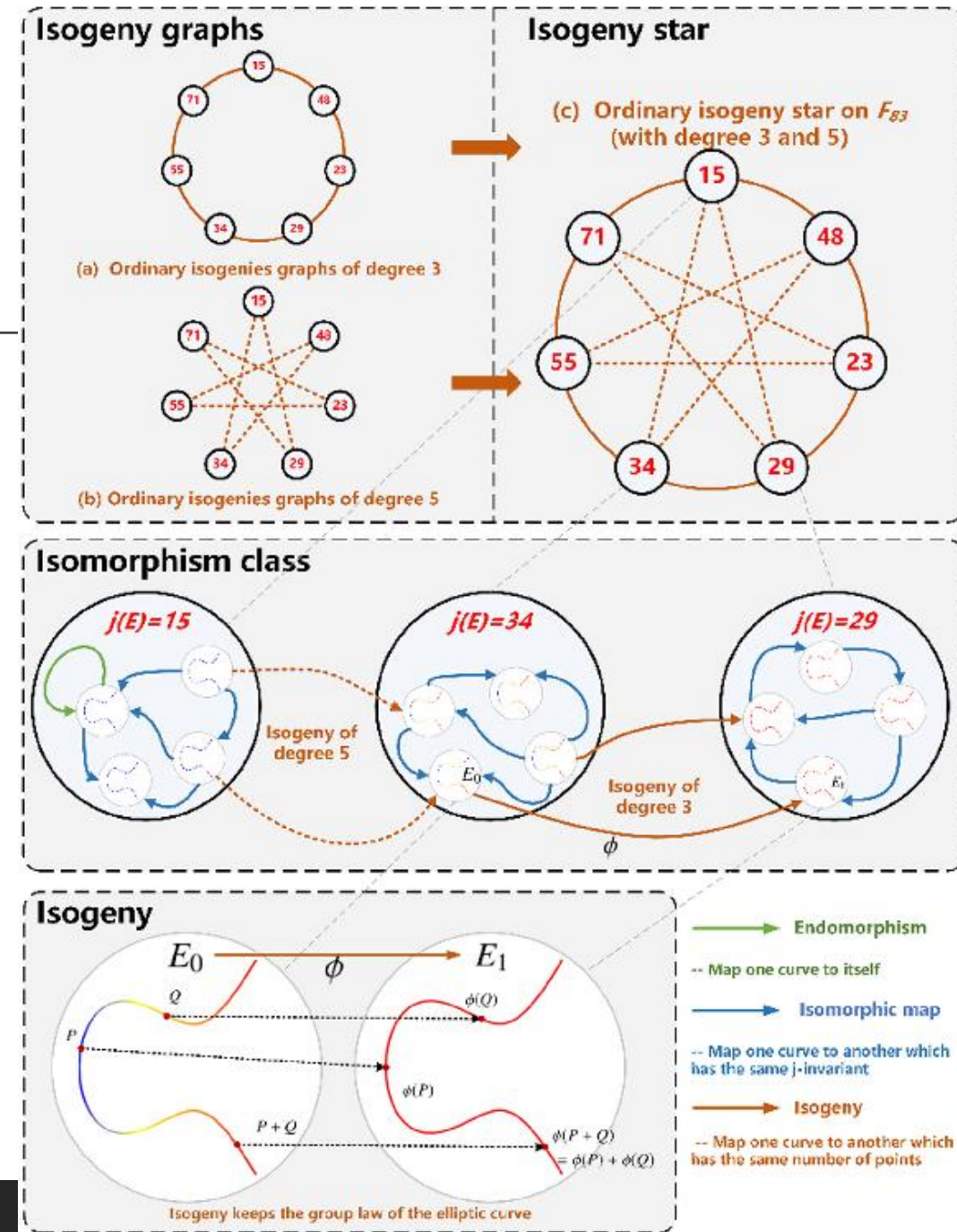
Multivariate Cryptography

- The design of cryptography based on **multivariate (quadratic) polynomial**.
- The underlying security lies on the difficulty of **solving multivariate quadratic polynomial over finite field**.



Multivariate Cryptography

- The youngest candidate in the PQC.
- The design of cryptography based on **elliptic curve cryptography**.
- The underlying security lies on the difficulty of **finding an isogeny star**.



~ THE END 😊 ~